

Cross-Prime Gram Coherence in a Literal Twin-Prime Dynamics: A Finite Phase Diagram and Sign-Flip Obstruction

Liang Wang

School of Mathematics and Statistics
Huazhong University of Science and Technology (HUST)
Wuhan, China

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Abstract

The preceding finite shell study found a potentially misleading combination: a four-block scalar attachment can be small while the physical output energy of the prime shell is amplified. This paper follows the next structural question without changing the literal operator or source. For the physical prime-component output vectors g_q we form the source-native Gram matrix $G_{q,r} = \langle g_q, g_r \rangle$ and the normalized squared coherence $\Gamma_{q,r} = G_{q,r}^2 / (G_{q,q}G_{r,r})$. We prove exactly a conditional coherence-accumulation inequality: a positive coherence floor and diagonal balance force a lower bound on the aggregate energy ratio. An exact-rational finite scan of 18 rows and 1,380 unordered pairs then separates two regimes. Seventeen rows are pairwise positive, but the early $(N, H, Q, z, s) = (256, 38, 27, 5, 1)$ row contains three negative cross-prime pairs, one with squared coherence about 1.37×10^{-7} . Conversely, eight late-shell rows satisfy the finite thresholds $\Gamma_{q,r} \geq 9/25$ and $d_{\min}/d_{\max} \geq 4/5$ for every pair, while all 18 rows have aggregate energy ratio greater than one. The result is a finite sign/coherence phase diagram and a precise obstruction to promoting late-shell behavior to a uniform growing-shell theorem; it is not an asymptotic twin-prime result or an arithmetic L^2 estimate.

1 Position on the route map

The working line studies a single literal dynamical family intended to model a prime-shell contribution to the twin-prime problem. Earlier releases split the finite physical operator into prime components and documented scalar cancellation. The immediately preceding release, TPC-288, retained the full output vectors and proved the exact Gram identity while certifying full rank on a finite growth/control grid [1]. Its central warning was that a small scalar attachment is not an L^2 saving.

TPC-289 asks the smallest follow-up question suggested by that warning: whether the cross-prime terms themselves have a stable sign and size. This is more informative than another rank computation, but it remains deliberately finite. The distinction between exact identities, finite numerical certificates, and open asymptotic gates is maintained throughout.

2 The frozen physical model

Let $I = I_N$ be the integer interval used by the frozen source engine and let $\beta \in \mathbb{Q}^I$ be its source vector. For an odd prime q , define the centered residue kernel

$$B_q(u, t) = \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} \left(\mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right).$$

With

$$K_{H,s}(h) = \frac{H^{2s}}{(H^2 + h^2)^s}, \quad D_q(u, t) = \mathbf{1}_{u \neq t} B_q(u, t),$$

the literal deleted-diagonal component is

$$(A_q)_{u,t} = q K_{H,s}(u-t) D_q(u, t), \quad g_q = A_q \beta.$$

For a finite shell S write

$$g_S = \sum_{q \in S} g_q, \quad G_{q,r} = \langle g_q, g_r \rangle, \quad d_q = G_{q,q} = \|g_q\|_2^2.$$

No four-block scalar attachment is applied in this paper. The source, kernel, deleted diagonal, and prime shell are exactly those inherited from TPC-288.

For $q \neq r$ with $d_q d_r > 0$, define

$$\Gamma_{q,r} = \frac{G_{q,r}^2}{d_q d_r}.$$

The square records the size of a cross term independently of its sign; the sign of $G_{q,r}$ is retained as a separate datum. Expanding the aggregate square gives the exact energy decomposition

$$\frac{\|g_S\|_2^2}{\sum_{q \in S} d_q} = 1 + \frac{\sum_{q \neq r} G_{q,r}}{\sum_{q \in S} d_q}, \quad (1)$$

where the off-diagonal sum is over ordered pairs.

3 Exact coherence structure

Lemma 1 (Gram positivity and coherence bound). *For every finite shell, G is positive semidefinite. If $d_q, d_r > 0$, then*

$$0 \leq \Gamma_{q,r} \leq 1. \quad (2)$$

Proof. For any real coefficient vector c ,

$$c^T G c = \sum_{q,r} c_q c_r \langle g_q, g_r \rangle = \left\| \sum_q c_q g_q \right\|_2^2 \geq 0.$$

The lower bound in (2) is a square and the upper bound is Cauchy–Schwarz divided by $d_q d_r > 0$. $\square$

Proposition 1 (Conditional coherence accumulation). *Let $k = |S|$, $d_- = \min_{q \in S} d_q$, and $d_+ = \max_{q \in S} d_q$. Suppose that for some $0 \leq \eta, \delta \leq 1$,*

$$G_{q,r} > 0, \quad \Gamma_{q,r} \geq \eta^2 \quad (q \neq r), \quad \frac{d_-}{d_+} \geq \delta.$$

Then

$$\frac{\|g_S\|_2^2}{\sum_q d_q} \geq 1 + \eta \delta (k - 1). \quad (3)$$

Proof. The positivity assumption selects the positive square-root branch in the definition of $\Gamma_{q,r}$, so

$$G_{q,r} \geq \eta \sqrt{d_q d_r} \geq \eta d_-.$$

There are $k(k-1)$ ordered off-diagonal terms. Since $\sum_q d_q \leq k d_+$, equation (1) yields

$$R_E \geq 1 + \frac{k(k-1)\eta d_-}{k d_+} \geq 1 + \eta \delta (k - 1).$$

$\square$

Remark 1. *The proposition is a finite vector lemma, not a claim that its hypotheses hold for all prime shells. If even one cross term is negative, its hypotheses fail, irrespective of whether the total energy ratio happens to exceed one. This logical separation is the obstruction tested below.*

4 Finite protocol

The certificate uses exact Python ‘Fraction’ arithmetic and the frozen TPC-268 engine. The 18 rows consist of eight $s = 2$ growth anchors, four selected $s = 1$ exponent-crossover rows, and six $s = 2$ source-control rows at $Q = 70$. The shell anchors range from three to 17 primes. The six controls use $H \in \{48, 52\}$ and $z \in \{3, 5, 7\}$.

For every row the producer reconstructs each g_q , forms every rational Gram entry, checks (2), records the sign census, and computes the exact energy ratio in (1). The finite “strong block” test uses

$$\eta = \frac{3}{5}, \quad \delta = \frac{4}{5}, \quad \eta^2 = \frac{9}{25}.$$

Thus a 15-prime row passing the test has the conditional lower envelope $R_E \geq 1 + (3/5)(4/5)14 = 193/25$, while a 17-prime row has $R_E \geq 217/25$.

The producer locks the TPC-288 source code and result, as well as the frozen engine, by normalized-LF SHA-256. An independent checker reverses the output summation order and reconstructs the complete canonical JSON document. A separate stress script rejects ten mutations, including altered signs, thresholds, row counts, energy values, and provenance. Normal and optimized Python executions are required to have byte-identical output.

5 Results: a finite sign/coherence phase diagram

Table 1 gives representative rows. “min Γ ” is the smallest squared coherence over unordered pairs; “+/-” is the pair sign census; and “strong” means that all three hypotheses in (1) pass with the declared (η, δ) .

axis	N	H	Q	s	k	min Γ	+/-	R_E	strong
growth	128	24	9	2	3	0.292716	3/0	2.124905	no
growth	192	32	16	2	5	0.251582	10/0	3.443701	no
growth	256	38	27	2	7	0.000949	21/0	2.622578	no
growth	384	50	40	2	10	0.207063	45/0	7.106564	no
growth	512	58	60	2	13	0.331366	78/0	10.992677	no
growth	512	58	70	2	15	0.444464	105/0	13.558338	yes
growth	512	58	90	2	17	0.806422	136/0	16.439313	yes
cross	256	38	27	1	7	1.37×10^{-7}	18/3	2.166576	no
cross	384	50	40	1	10	0.032697	45/0	4.557433	no
cross	512	58	70	1	15	0.058904	105/0	8.900964	no
control	384	48	70	2	15	0.721183	105/0	14.322934	yes
control	384	52	70	2	15	0.642607	105/0	14.107249	yes

Table 1: Selected exact-rational rows; decimals are display strings only.

Across all 18 rows there are 1,380 unordered pair comparisons. Seventeen rows are pairwise positive and one row has three negative pairs; no pair is zero. Every row has $R_E > 1$. The exceptional row is (256, 38, 27, 5, 1), with shell {29, 31, 37, 41, 43, 47, 53}. Its negative pairs are

$$(29, 53), \quad (31, 53), \quad (41, 53).$$

The squared coherence of (31, 53) is approximately 1.3745664×10^{-7} , while (29, 53) has negative Gram cross term and squared coherence approximately 0.0105163. Thus the failure is not merely a rounding ambiguity: the signs are exact rational signs and one pair is nearly orthogonal.

The late block consists of eight rows: the $Q = 70$ and $Q = 90$ $s = 2$ growth anchors, together with the six $Q = 70$ controls. Every pair is positive, every squared coherence is at least $9/25$, and the diagonal ratio is at least $4/5$. The conditional proposition therefore applies row by row. The finite certificate also finds two exact groups in which the three cutoff controls at a fixed height have the same recorded coherence/energy signature; this is a warning that these finite controls should not be counted as an asymptotically independent source family.

6 What is and is not established

The positive late block is useful because it supplies a quantitative mechanism for the energy amplification seen in TPC-288: coherent cross terms accumulate linearly in the number of components. The early sign flip is equally useful: it refutes the tempting uniform rule

“all declared shell cross terms are positive and have a fixed coherence floor.”

The observed total $R_E > 1$ in the exceptional row shows that aggregate amplification does not require pairwise positivity either.

The result is therefore an obstruction paper on the route map, not a proof or disproof of the twin-prime conjecture. In particular, we do not claim a growing-shell bound, a uniform theorem over literal sources, an arithmetic L^2 estimate, a fixed-power credit, or a Gate-B conclusion. The next minimal questions are whether an adaptive shell weighting can exploit the coherence without violating the physical operator, and whether a restricted source class admits a uniform coherence estimate.

7 Reproducibility and claim firewall

The complete artifact is in [the project README](#). The canonical result is ‘results/tpc289_certificate.json’; the proof and derivation packages state the exact scope. The local fallback evaluator reports:

```
EXACT: Gram PSD, 0 <= Gamma <= 1, conditional accumulation envelope
FINITE: 17/18 positive rows; 3 negative pairs in one row
FINITE: 8-row eta=3/5, delta=4/5 strong block; 18/18 R_E > 1
OPEN: source-restricted/growing-shell coherence and arithmetic L2
NONE: twin-prime conclusion
```

The Session-named Route-A/Route-B evaluator files are not present in this checkout, so no official evaluator pass is claimed. The independent replay, mutation audit, and bridge checker are the available fail-closed evidence.

References

- [1] Liang Wang. Growing-shell gram obstruction, 2026. TPC-288 project artifact in the prime dynamics theory session.